Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :- Reduction of breakdowns in W shop due to kickless cable failure

Mahindra and Mahindra Ltd, Plant I, Scorpio Body shop.

	Plan	Actual
Start Date	**25 Aug 12**	**30 Aug 12**
Completion Date	**25 Nov12**	**24 Nov 12**

Current Status:

Closed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Vinod Autti	Dy. Manger	SBS	9922448558	
2	**Mentor**	H S Nagar	Manager	SBS	9881786767	
3	**Member 1**	Sahdeo Masurkar	Asst. Manger	SBS	9766140001	
4	**Member 2**	Kiran Deshmukh	Asst. Manger	SBS	8308833228	
5	**Member 3**	Rajendra Punde	Asst. Manger	SBS	9922228599	
6	**Member 4**	Sunil Deshmukh	Asst. Manger	SBS	8380077969	
7	**Member 5**	Shailesh kulkarni	Asst. Manger	SBS	9370843431	

1. Select the Theme

Brief description of Problem

What: Kickless cable failure

Where: W shop (All Spot welding guns)

When : Daily(1 & 2 shift)

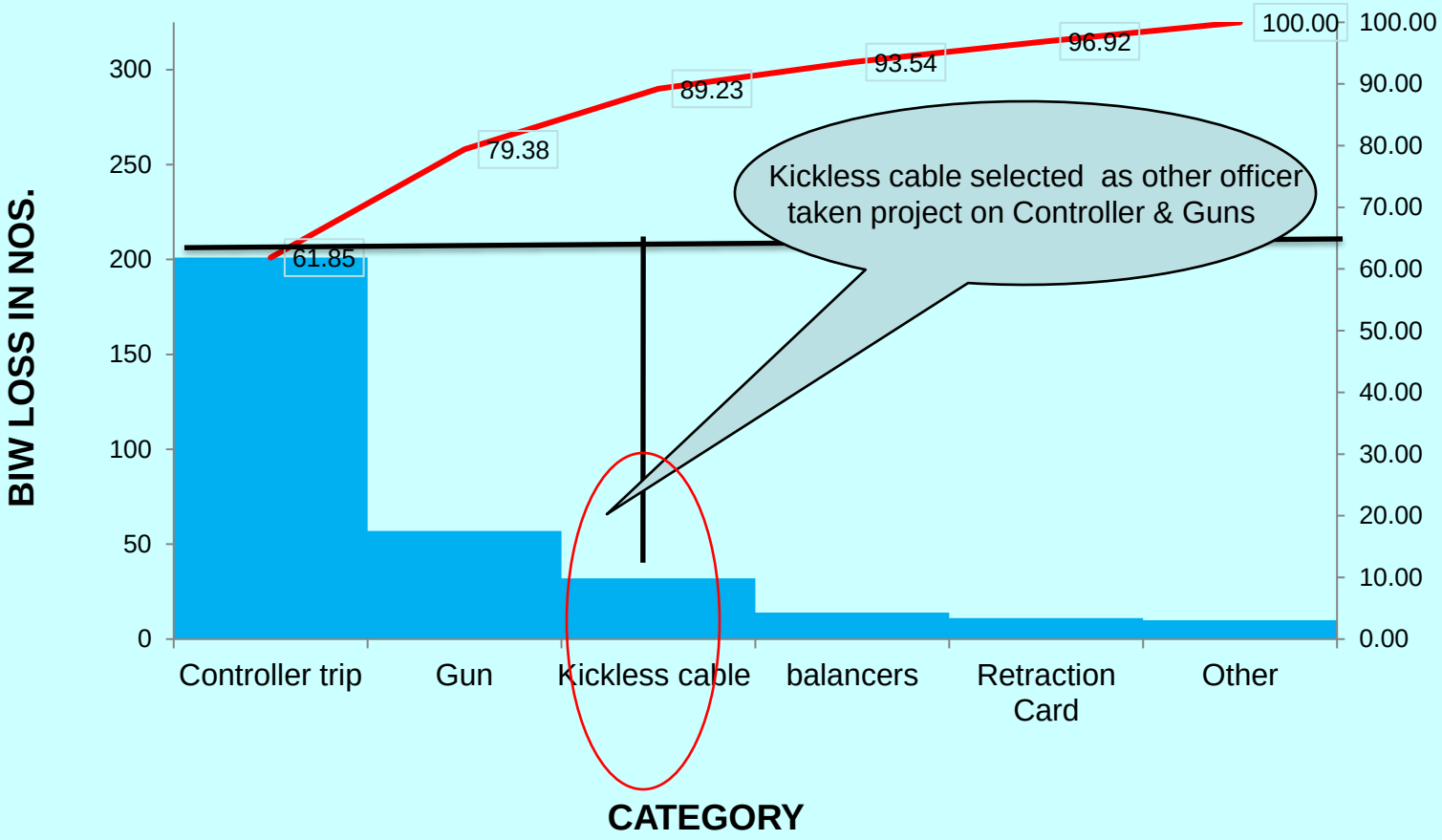
Who : Maintenance officer & production officer

Why : High MTTR & production loss

How: Average @ 12 incidences reported per month

2. Justify the choice (reasons)

F12 BIW loss on spot welding



2.a Immediate Containment Action

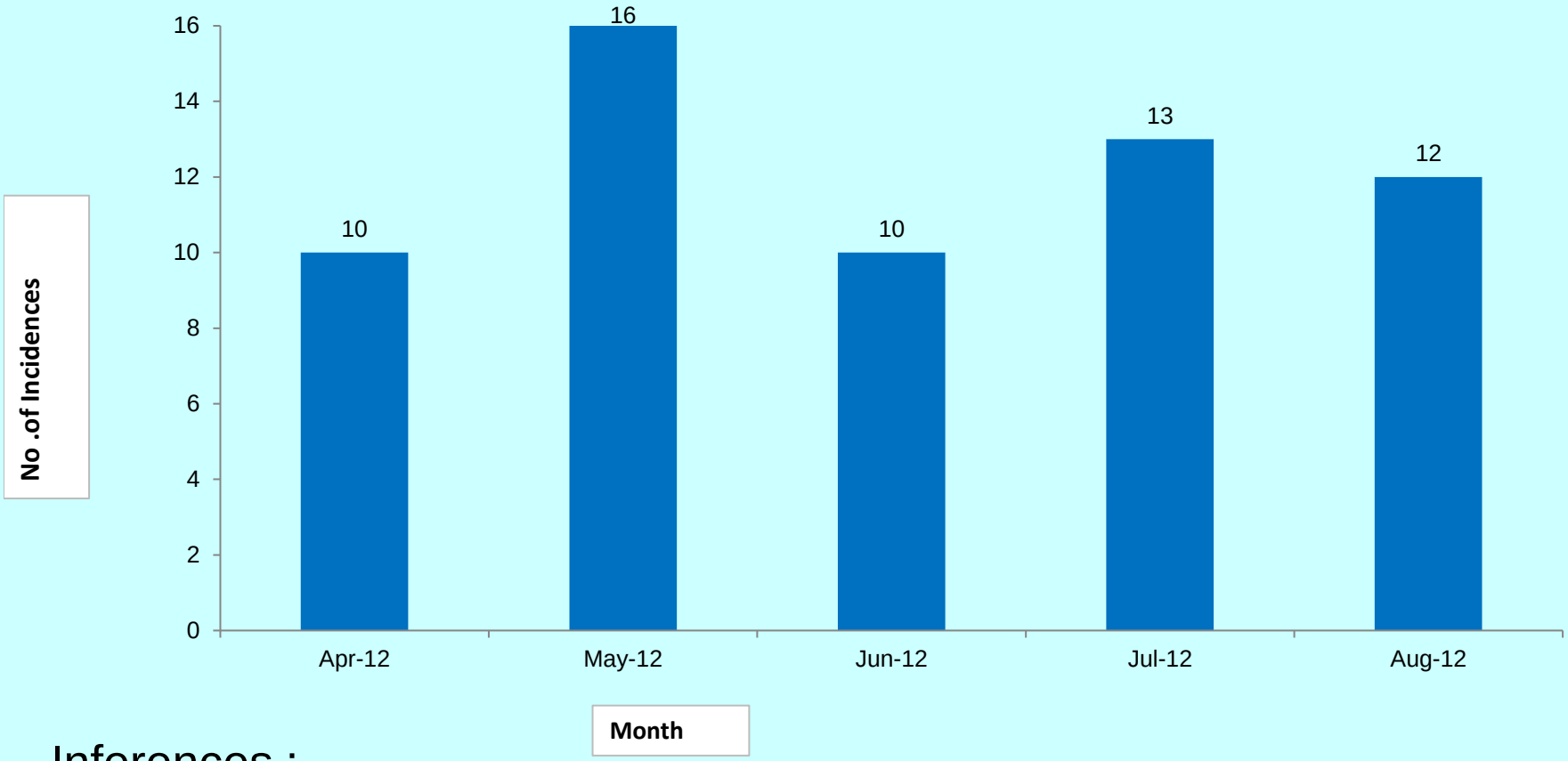
- Measuring Actual current Vs Set current.
- Measuring resistance of kickless cable.
- Kickless cable replacement .

3. Understand the current situation

Application of kickless cable

Kickless cable is use to transfer welding current from welding transformer to spot welding gun

3. Understand the current situation

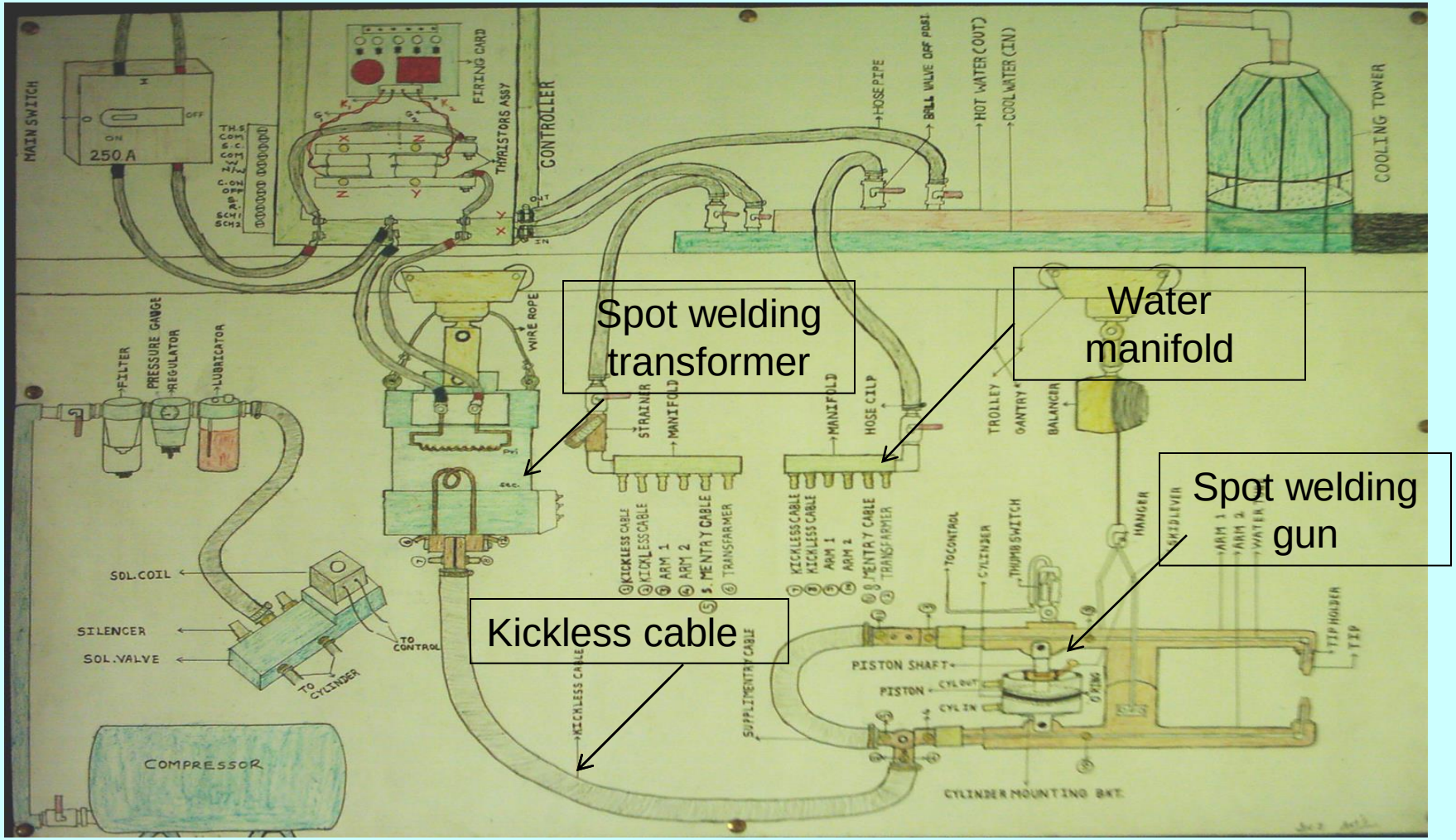


Inferences :

1. Kickless Cable Cost Avg. Rs 10,000 / Cable
2. Cost of poor Quality Avg. Rs 41,000 / BIW

3. Understand the current situation

Spot Welding gun Schematic



3. Understand the current situation

Spare parts of Kickless Cable

Copper Conductor Bunch

Bakelite Insulators (separators)

Copper End Connectors

Rubber Star Separator

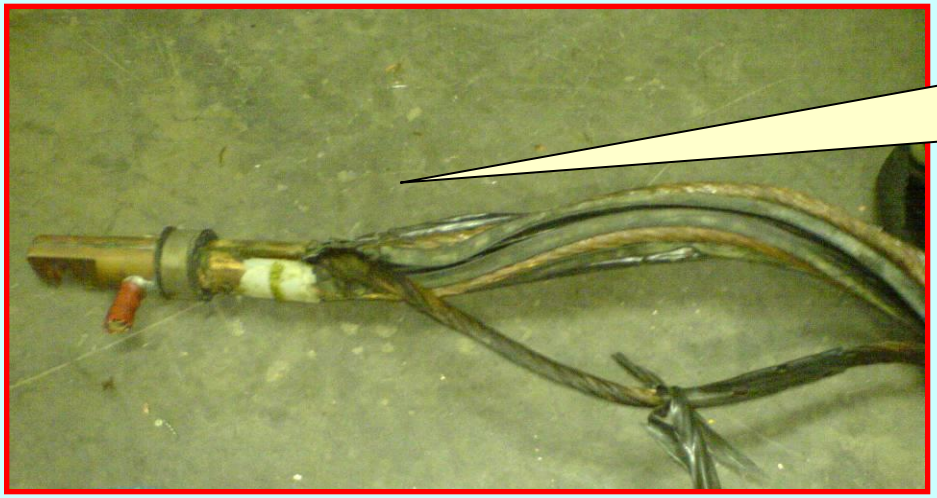
Rubber Hose Pipe

Single Copper Conductor

3. Understand the current situation

Kickless cable Lengthwise population		
Sr No	Kickless Cable length	Population
1	Kickless Cable 3M	77
2	Kickless Cable 2.5M	44
3	Kickless Cable 3.5M	22
4	Kickless Cable 2 M	14
5	Kickless Cable 4M	8

3. Understand the current situation



Faulty portion of kickless cable

Damaged copper conductors

4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Average 12 no's kickless cable failure /month
Target:	80% Reduction in kickless cable breakdowns

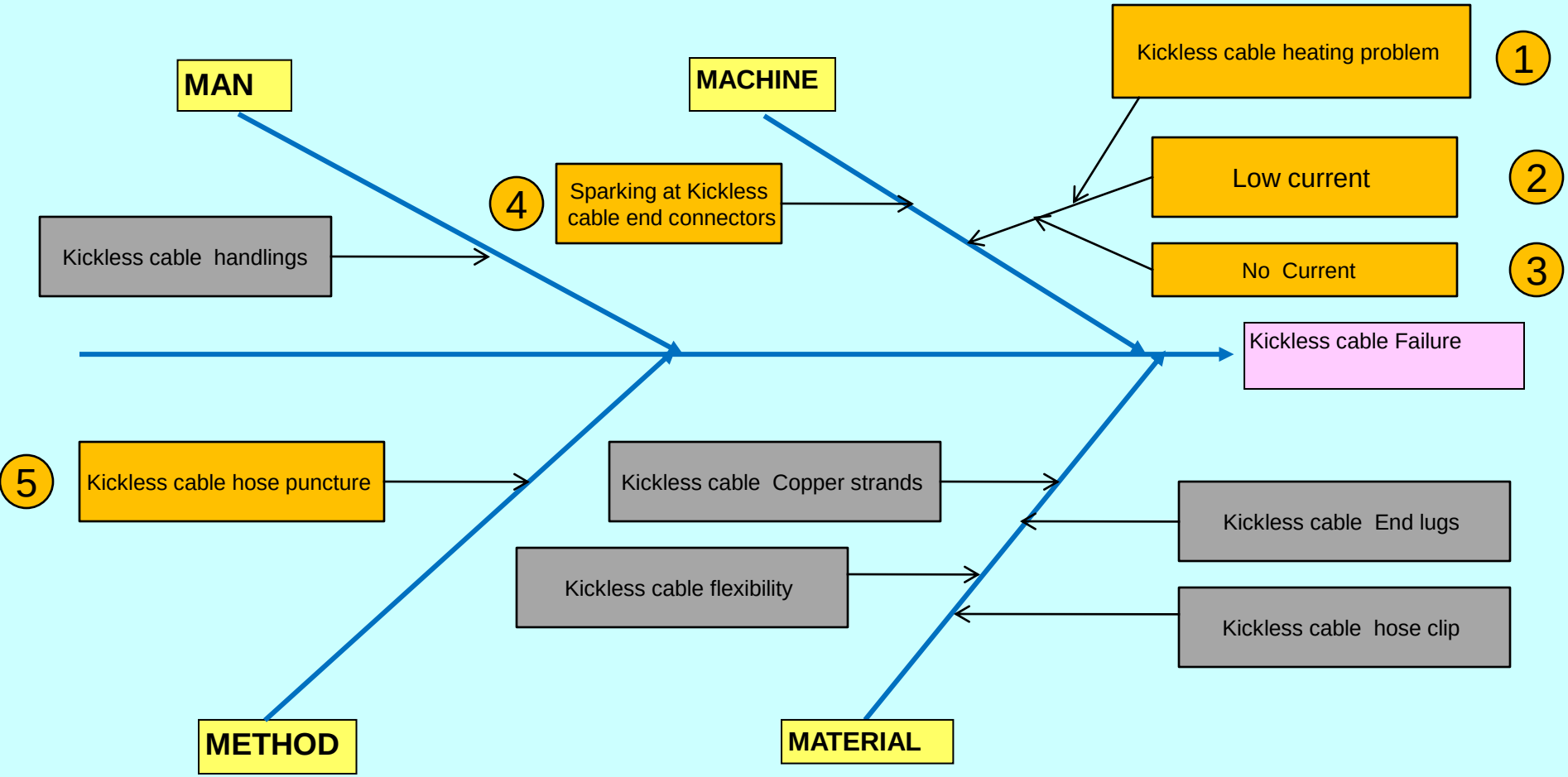
5. Analysis

Brain storming by CFT:- Possible causes

Sr. No.	Possible Causes
1)	Kickless cable heating problem
2)	Low current
3)	No Current
4)	Kickless cable hose puncture
5)	Sparking at Kickless cable end connectors.
6)	Kickless cable flexibility
7)	Kickless cable hose clip
8)	Kickless cable end lug
9)	Kickless Cable Copper strands
10)	Kickless cable handling

5. Analysis

Cause & Effect Diagram :



5.1 Test Of Hypothesis

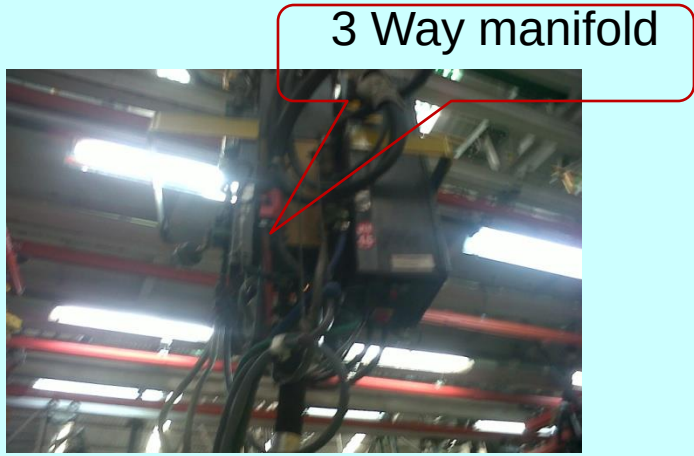
Probable causes

Sr. No.	Probable Causes
1)	Kickless cable heating problem
2)	Low current
3)	No Current
4)	Kickless cable hose puncture
5)	Sparking at Kickless cable end connectors.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1)	Kickless cable heating due to inadequate flow of process water	Water flow measurement through Rota meter found process water Flow less then 7 LPM also strainer found choke as there is no process water filter in system also for british federal transformer 3 way manifold is used due to cooling water circulation system is in series which should be in parallel	**Hypothesis Valid**



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2)	Low current	Spot welding secondary current measured by using weld checker current found 7.5 KA against specified 8.5 KA	**Hypothesis Valid**

Transformer No :19

Sr no	Set Current (in KA)	Actual Current (in KA)
1	8.5	8.1
2	8.5	8.5
3	8.5	8.6
4	8.5	8
5	8.5	7.8
5	8.5	7.8
6	8.5	7.8
7	8.5	7.9
8	8.5	7.9
9	8.5	7.9
10	8.5	7.8

Tolerance +/-15%

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
3)	No Current	Spot welding secondary current measured by using weld checker current found 8.5 KA against specified 8.5 KA	**Hypothesis Invalid**

Transformer No :59

Sr no	Set Current (in KA)	Actual Current (in KA)
1	9.3	9.2
2	9.3	9.2
3	9.3	9.3
4	9.3	9.3
5	9.3	9.2
5	9.3	9.2
6	9.3	9.3
7	9.3	9.2
8	9.3	9.2
9	9.3	9.3
10	9.3	9.3

Tolerance +/-15%

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

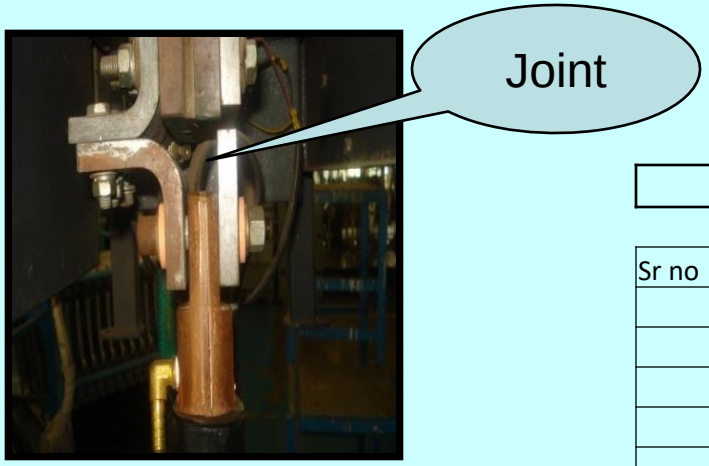
No	Probable Cause	Testing and Observations	Conclusion
4)	Kickless cable hose puncture	Visually observed kickless cable rubber hose touching metallic hanger body while spotting resulting in friction which should not touch while taking spot	**Hypothesis Valid**

Kickless cable hose Touching hanger

TEST OF HYPOTHESIS

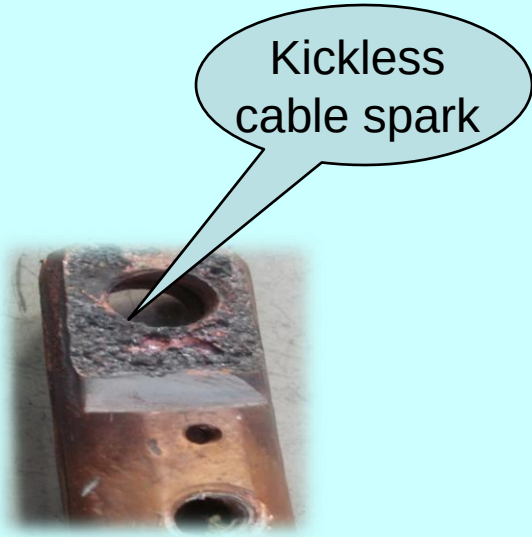
5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5)	Sparking at Kickless cable end connectors due to loose connection at joint	Visually Tested by using spanners found for two transformers loose connection	**Hypothesis Valid**



Kickless cable end connection Monitoring

Sr no	Transfomer	gun end	Transformer end
1	59	ok	ok
2	118	ok	ok
3	75	ok	ok
4	90	ok	ok
5	95	Not ok	ok
6	8	ok	Not ok
7	34	ok	ok
8	35	ok	ok
9	111	ok	ok
10	85	ok	ok



WHY - WHY Analysis

5.3 Identify the root causes: (Use Format MQS/MAPS/F05)

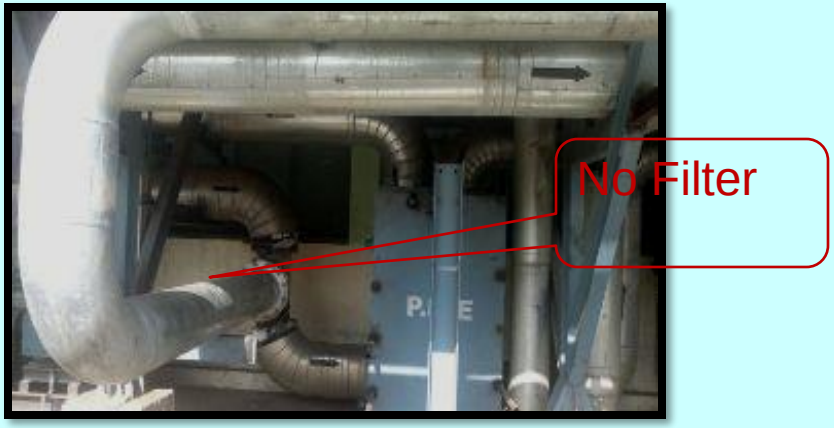
Valid Probable Cause	WHY	WHY	WHY	WHY	WHY
Kickless cable heating	Insufficient water flow through kickless cable	Y Strainer choke up	No filter in process water pipe line		
Low current	Kickless cable stands weak	Ageing of kickless cable			
Water leakage through kickless cable	Kickless cable hose puncture	Kickless cable hose in contact with hanger while spotting			
Sparking at Kickless cable end connectors	Loose connection at gun & transformer end	Frequent movement of gun & Transformer			

6. Implement the corrective measures

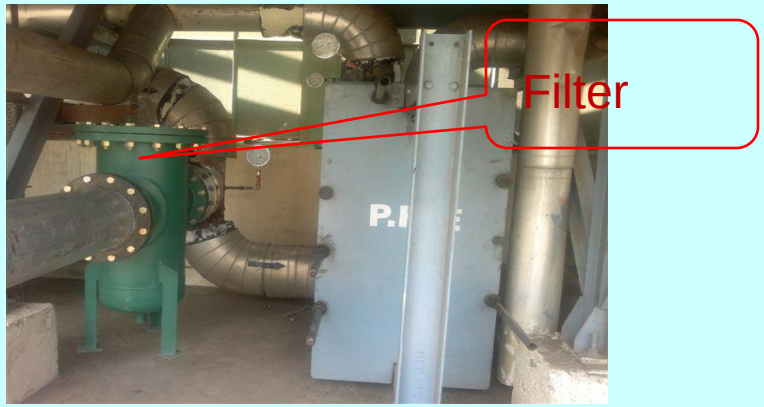
6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Kickless cable heating problem	Process water filter provided & 10 way manifold in place of 3 way for british federal transformer with higher spots	13.11.12	Vinod	Closed
Low current	Conditional based Monitoring for kickless cable	12.11.12	Sahdeo	Closed
Water leakage through kickless cable	Protective cover provided to avoid friction	19.11.12	Sunil	Closed
Sparking at Kickless cable end connectors	Kickless cable tightening theme schedule	15.11.12	Vinod	Closed

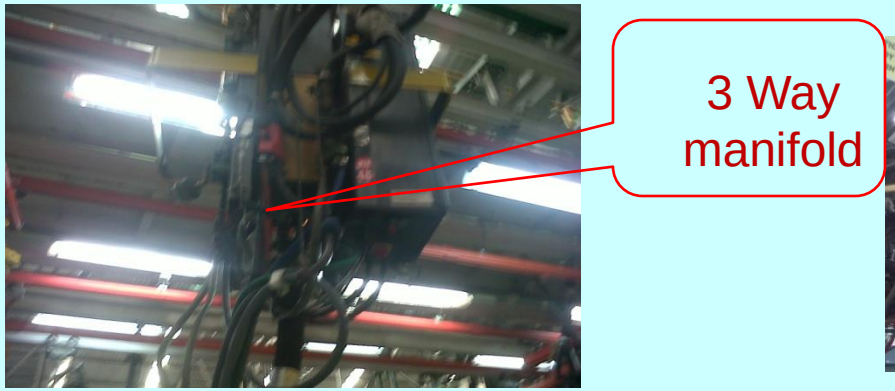
6. Implement the corrective measures



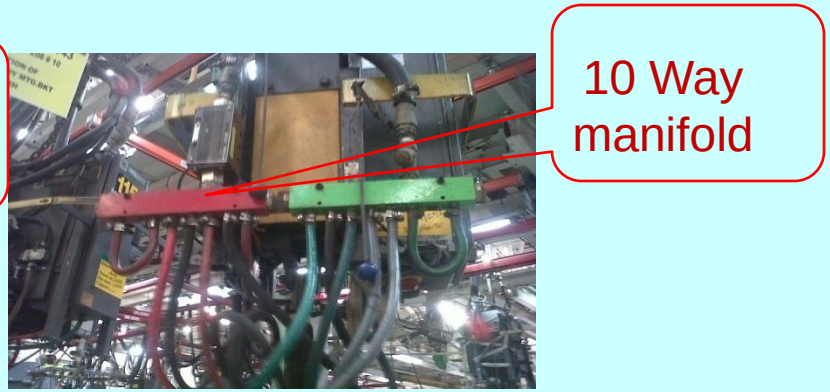
No process water filter



process water filter



3 Way manifold



10 Way manifold

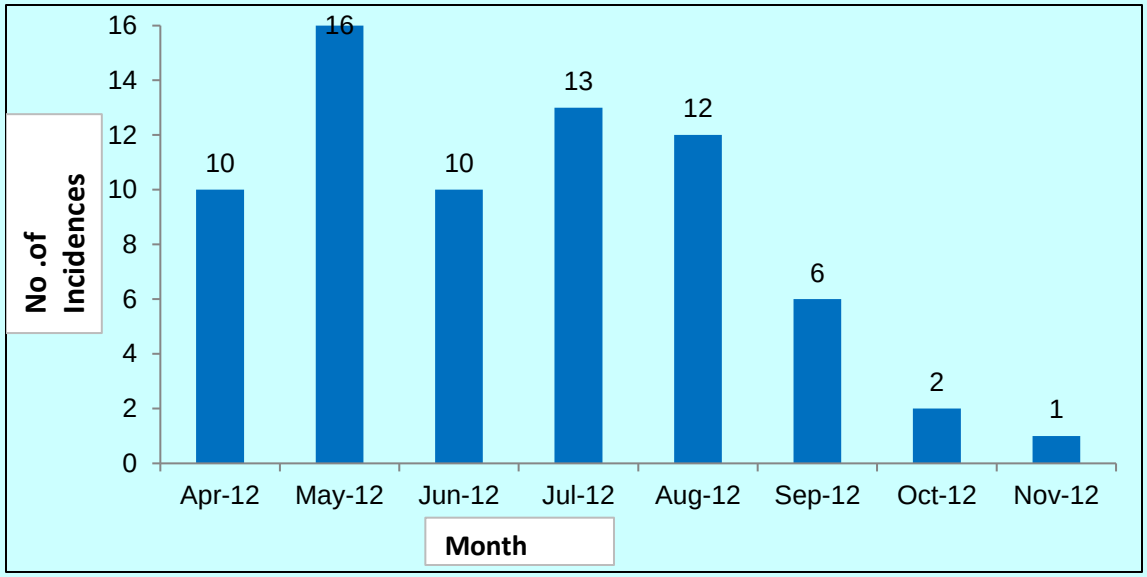
6. Implement the corrective measures

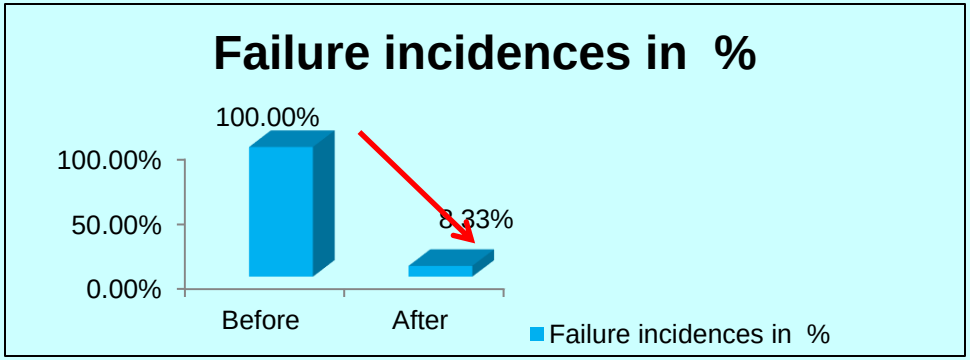
Hose friction with hanger during spotting

Protective cover provided to prevent friction

7. Check & confirm the results & effects

7.1 Verify Results:





• 7.2 Full Scale Implementation plan:

1. All decided actions implemented.

2. Horizontal deployment Plan for SCDC Shop

Check & confirm the results & effects (Cont……)

7.3 Tangible :

1. Kickless Cable Cost saved Rs 90,000/annum
2. Cost of Poor Quality Saved Rs 41,000/ Failure
3. Productivity improved as there is 80% reduction in kickless cable failure

7.3 Intangible benefits:

1. Team Morale improved
2. Team Work Built up
3. Team Confidence Boost up
4. Knowledge of Worker improved

8. Standardize

8.1 Updation of documents:

Sr No	Document	Target Date	Responsibilty	Status
1	Kickless cable end connection lugs tightening Theme in Preventive maintenance	Sep-12	Hari Nagar	Closed
2	Spot Welding Gun Y Strainer cleaning Theme in preventive maintenance	Sep-12	Hari Nagar	Closed

8.2 Training :

Sr No	Training Identified	Target Date	Responsibilty	Status
1	Training Given to operators for kickless cable parts & its function	Nov-12	Sahdeo Masurkar	Closed

9. Summarise & Plan future actions

9.1 Conclusion:

An innovative approach and team work by doing GEMBA study can lead to elimination of such chronic concerns

9.2 Horizontal Deployment:

Project initiated for SCDC spot welding guns will be implemented by Jan13

9.3 Lessons Learnt:

Things Gone Wrong :

Kickless cable outlet water temperature checking not planned

Things Gone Right :

We study the process of spot welding at different stages & identified different causes of kickless cable failure also we studied that Problems can be easily solved by using QC story approach

8.3 Additional Audit / Check

1. PM Check Sheet (kickless cable tightening, strainer cleaning)

Thank You……MQS TEAM